PRODUCT OVERVIEW & MACHINE LEARNING MODEL SPECIFICATIONS

1. Introduction, Purpose and Scope

This Software Requirements Specification (SRS) details the operational, functional, architectural, and mathematical requirements for the **Handwritten Digit Recognition System**. The application delivers real-time inference of isolated handwritten numerical digits (*0* through *9*) using a Convolutional Neural Network (CNN) integrated into a Python Flask web service with client-side canvas interactions, persistent SQLite storage, and analytics visualization.

2. Overall System Description & Operational Context

The system operates as a self-contained, three-tier web application:

- **Client Tier:** HTML5 Canvas (high-DPI, touch/stylus/mouse enabled), Bootstrap 5, Chart.js 4.
- **Application Tier:** Python 3.12 Flask REST server executing image transformation, preprocessing, and TensorFlow/Keras inference.
- **Data Tier:** Embedded SQLite 3 (`history.db`) for transaction logging and `digit_model.keras` for model weights.

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3. Machine Learning & Deep Learning Model Specifications



3.1 Model Topology & Parameter Derivation

The classification engine employs a sequential Convolutional Neural Network containing **225,034 total trainable parameters** (zero non-trainable parameters) across the following topological layers:

Layer Identifier	Layer Type	Output Shape	Kernel / Receptive Field	Activation	Parameter Calculation	Trainable Params
`digit_input`	InputLayer	*(28, 28, 1)*	—	—	*0*	*0*
`conv_1`	Conv2D	*(26, 26, 32)*	*3 × 3*, Stride 1, Valid	ReLU	*(3 × 3 × 1 + 1) × 32*	**320**
`pool_1`	MaxPooling2D	*(13, 13, 32)*	*2 × 2*, Stride 2	—	None	*0*
`conv_2`	Conv2D	*(11, 11, 64)*	*3 × 3*, Stride 1, Valid	ReLU	*(3 × 3 × 32 + 1) × 64*	**18,496**
`pool_2`	MaxPooling2D	*(5, 5, 64)*	*2 × 2*, Stride 2	—	None	*0*
`dropout_conv`	Dropout	*(5, 5, 64)*	Rate = *0.25*	—	None	*0*
`flatten`	Flatten	*(1600)*	*5 × 5 × 64* units	—	None	*0*
`dense_hidden`	Dense	*(128)*	Fully connected	ReLU	*(1600 + 1) × 128*	**204,928**
`dropout_dense`	Dropout	*(128)*	Rate = *0.50*	—	None	*0*
`digit_probs`	Dense	*(10)*	10 output classes	Softmax	*(128 + 1) × 10*	**1,290**
TOTAL	—	—	—	—	**All Trainable**	**225,034**

3.2 Mathematical Formulation of Output Probabilities

Logits z_k produced by `digit_probs` are normalized via the categorical Softmax activation function:

> $$p_k = \sigma(z)_k = (e^{z_k}) / (\sum_{j=0}^{9} e^{z_j}), \quad \forall k \in \{0, 1, \dots, 9\}, \quad \text{such that} \quad \sum_{k=0}^{9} p_k = 1.0$$

The predicted digit is determined by the maximum likelihood class: $\hat{y} = \arg\max_{k \in \{0, \dots, 9\}} p_k$, with confidence $c = \max_k p_k$.

3.3 Training Strategy & Optimization Configuration

- **Dataset:** Modified NIST (MNIST) consisting of *60,000* training images and *10,000* test images.
- **Loss Function:** Sparse Categorical Cross-Entropy: $\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} \log(\sigma(\mathbf{z}^{(i)})_{y^{(i)}})$.
- **Optimizer:** Adam ($\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$).
- **Batch Size & Epochs:** Mini-batch size of *128* over *6* full training epochs.

FUNCTIONAL, INTERFACE, AND NON-FUNCTIONAL REQUIREMENTS

4. Functional Requirements Specifications

`User` → `FR1 & FR2 & FR3 & FR4 & FR5 & FR6 & FR7 & FR8`

- **FR-1: Multi-Modal Canvas:** The system shall support mouse, touch, and stylus input on a $280 \times 280\text{px}$ canvas, scaling client coordinates accurately while applying `touch-action: none` to block scrolling. It shall support Brush and Eraser tools (4px–32px), multi-step Undo/Redo, and keyboard shortcuts (`Ctrl+Z`, `Ctrl+Y`, `C`, `P`, `B`, `E`).
- **FR-2: Smart & Auto Prediction:** The system shall support on-demand inference and an optional ~700ms debounced auto-predict mode triggered after stroke completion. Repeated blank submissions shall be rejected with `HTTP 400`.
- **FR-3: Probability Visualization:** The system shall display the predicted digit, confidence percentage, top 3 candidates, and a full 10-class horizontal bar chart rendered via Chart.js highlighting the winning digit.
- **FR-4: Preprocessing Inspector:** The system shall provide an expandable inspection card illustrating original input, polarity-corrected grayscale, and final 28×28 centered tensor image.
- **FR-5: History Persistence (SQLite):** All predictions shall persist across page reloads and server restarts in SQLite (`history.db`). The UI shall provide real-time search, filtering by digit (*0–9*) and modality (*Canvas*, *Upload*, *Sample*), sorting by date and confidence, and single/bulk deletion.
- **FR-6: Safe CSV Export:** The system shall export historical records via `GET /api/history/export` with UTF-8 BOM encoding and prefix sanitization (`'`) for cells starting with `=`, `+`, `-`, `@`, `\t`, or `\r`.
- **FR-7: Verification Feedback Loop:** Users shall be able to mark predictions as "Correct" or "Incorrect" (with ground-truth actual digit selection), recorded in SQLite to track observed accuracy via `GET /api/feedback/stats`.
- **FR-8: Digit Challenge Mode:** The system shall generate a random target digit (*0–9*) and update a persistent player scoreboard (+10 pts per match, total attempts, running accuracy).

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5. External Interface & REST API Specifications

Method	Endpoint	Input Payload / Parameters	Success Output (JSON / Media)
`GET`	`/`	None	`200 OK` (Renders `index.html` single-page application)
`GET`	`/model-info`	None	`200 OK` (Renders `model_info.html` architectural report)
`GET`	`/health`	None	`{"status": "ok", "model_loaded": true, "version": "1.0.0"}`
`POST`	`/predict`	Multipart file (`file`) OR JSON `{"image": "..."}`	`{"success": true, "predicted_digit": int, "confidence": float, "probabilities": dict, "top_three": list, "preprocessing": dict, "prediction_id": str}`
`GET`	`/api/history`	Query params: `search`, `digit`, `input_type`, `sort_by`, `sort_order`	`{"success": true, "history": list, "total": int}`
`DELETE`	`/api/history/<id>`	Record ID in URL	`{"success": true, "message": "Record deleted"}`
`DELETE`	`/api/history`	None	`{"success": true, "message": "All history cleared"}`
`GET`	`/api/history/export`	None	`200 OK` (`Content-Type: text/csv; charset=utf-8`, attachment)
`POST`	`/api/feedback`	`{"prediction_id": str, "feedback": "correct"	"incorrect", "actual_digit": int}`
`GET`	`/api/feedback/stats`	None	`{"success": true, "stats": {"total_predictions": int, "total_feedback": int, "observed_accuracy": float}}`

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6. Non-Functional & Operational Requirements (NFR)

- **NFR-1: Latency & Performance:** Model inference execution shall complete within $\leq 30\text{ms}$ on standard CPU hardware. Total end-